

Notes ^{on} EDMD for stochastic systems

In stochastic systems, we typically have a discrete spectrum (eigenvalues) of the Koopmans operator defined as

$$U^t f(x_0) = \int P_t(x, x_0) f(x) dx \quad (1)$$

Here x_0 is some point in $\mathbb{R}^d$.

$P_t(x, x_0)$: transition probability density

read this as the conditional density of X_t given $X_0 = x_0$ evaluated at $X_t = x$.

When we have a deterministic system, $P_t(x, x_0)$ does not exist.

Instead we get a delta distribution for $P_t(x, x_0) dx \equiv \delta_{\varphi^t(x_0)}(x)$

Thus, we can recover the definition

of the deterministic Koopman operator

$$\mathcal{K}^t f(x_0) = \langle f, \delta_{\varphi^t x_0} \rangle$$

$$= f(\varphi^t x_0).$$

U^t is called a stochastic Koopman operator or just Koopman operator and represents conditional expectations of observables after time t given fixed values of x at the current time.

$$U^t f(x_0) = \mathbb{E}_{W \sim P} f \circ \varphi_W^t(x_0)$$

where the expectation is over the Wiener measure.

The adjoint of the Koopman semigroup (U^t satisfies the semigroup property $U^{t+s} = U^t \circ U^s$) is

the Frobenius-Perron or transfer operator semigroup, which gives the evolution of densities / probability distributions

$$\langle U^t f, \rho \rangle = \int f(x) \int p_t(x, x_0) \rho(x_0) dx_0 dx$$

$$= \int f(x) U^{*t} \rho(x) dx$$

$$= \langle f, U^{*t} \rho \rangle$$

$$\text{where } U^{*t} \rho(x) = \int p_t(x, x_0) \rho(x_0) dx_0$$

That is, if ρ is the density at the current time, $U^{*t} \rho$ is the density of X_t after time t .